Figure 4: Macroevolutionary Synteny Architecture and Codon Divergence Across Vertebrates

A. Calibrated Vertebrate Tree & Evolutionary Milestones

B. Conserved Syntenic Blocks & Chromosomal Architecture

C. Codon Decomposition (GC1, GC2, GC3) & Divergence

Agnatha
Chondrichthyes
Actinopterygii
Sarcopterygii
Amphibia
Archosauromorpha
Squamata (Inverted)
Mammalia (Canonical)

N1
N2
N3
N4
N5
N6
N7
N8

Sea Lamprey (P. marinus)
Grey Reef Shark (C. amblyrhynchos)
Catshark (S. canicula)
Spotted Gar (L. oculatus)
Zebrafish (D. rerio)
Fugu (T. rubripes)
Coelacanth (L. chalumnae)
Clawed Frog (X. tropicalis)
Axolotl (A. mexicanum)
Green Sea Turtle (C. mydas)
American Alligator (A. mississippiensis)
Chicken (G. gallus)
Anna's Hummingbird (C. anna)
Wall Lizard (P. muralis)
Corn Snake (P. guttatus)
Common Adder (V. berus)
Platypus (O. anatinus)
Gray Opossum (M. domestica)
Human (H. sapiens)
House Mouse (M. musculus)
Cattle (B. taurus)
Sperm Whale (P. catodon)

Proto-Hb
Ancestral Agnathan Single Locus
α
β
NPRL3
NPRL3-RHBDF1
◀ NPRL3-MPG-RHBDF1 (linked)
◀ NPRL3-MPG (linked)
OR51
Microchromosomes (Chr 14 & 25)
CDCA8-OR51-RRM1 ◀ INVERSION
NPRL3
NPRL3-MPG-LUC7L
FOLR1-OR51-NUMA1

ΔGC3: -8.8% (β > α)
ΔGC3: -7.1% (β > α)
ΔGC3: -5.9% (β > α)
ΔGC3: +13.2% (α > β)
ΔGC3: -23.7% (β > α)
ΔGC3: -22.7% (β > α)
ΔGC3: +8.3% (α > β)
ΔGC3: +7.6% (α > β)
ΔGC3: +25.3% (α > β)
ΔGC3: +6.6% (α > β)
ΔGC3: +24.0% (α > β)
ΔGC3: +20.4% (α > β)

500 Ma
400 Ma
300 Ma
200 Ma
100 Ma
0 Ma
600
500
400
300
200
100
0
−100
−200
Divergence Time (Million Years Ago - Ma)

5' Telomere
3' Telomere
0.0
0.2
0.4
0.6
0.8
1.0
Relative Chromosome Position (0.0 = 5' Telomere, 1.0 = 3' Telomere)

α-Globin GC1
α-Globin GC2
α-Globin GC3 (Synonymous Surge)
β-Globin GC1
β-Globin GC2
β-Globin GC3 (Synonymous Surge)
40
50
60
70
80
90
100
GC Content (%) [Diamond = GC3; Shaded Bar = GC12 to GC3 Surge]